

Vulnerability Assessment Report

1. Introduction

Vulnerability assessment is an important part of cybersecurity as it helps identify weaknesses in systems, services, and applications before they can be exploited by attackers.

In this task, a controlled test network was created using Kali Linux as the assessment machine and Metasploitable as the intentionally vulnerable target. Network reconnaissance, service enumeration, vulnerability scanning, and manual verification were performed to identify security weaknesses in the target system.

The assessment was conducted only against the intentionally vulnerable Metasploitable virtual machine in an isolated lab environment.

2. Objective

The objectives of this task were to:

- Identify the target system and verify network connectivity.
- Discover running services and their versions.
- Identify known vulnerabilities in exposed services.
- Analyze the severity and potential impact of identified vulnerabilities.
- Perform limited manual verification of selected findings.
- Document the vulnerabilities and provide appropriate mitigation recommendations.

3. Test Environment

Component	Details
Assessment Machine	Kali Linux
Kali IP Address	192.168.47.132
Target Machine	Metasploitable
Target IP Address	192.168.47.133
Virtualization	VMware
Network Type	Host-only / isolated lab network
Primary Tool	Nmap
Additional Tools	cURL
Target Purpose	Intentionally vulnerable test machine

The two virtual machines were placed on the same 192.168.47.0/24 network.

Connectivity was verified using ICMP before beginning the assessment.

4. Methodology

The assessment was performed in the following stages:

4.1 Network Connectivity Verification

The target was first tested using the `ping` command to verify that the Kali machine could communicate with the Metasploitable machine.

```
ping -c 4 192.168.47.133
```

The target responded successfully, confirming network connectivity.

4.2 Service and Version Enumeration

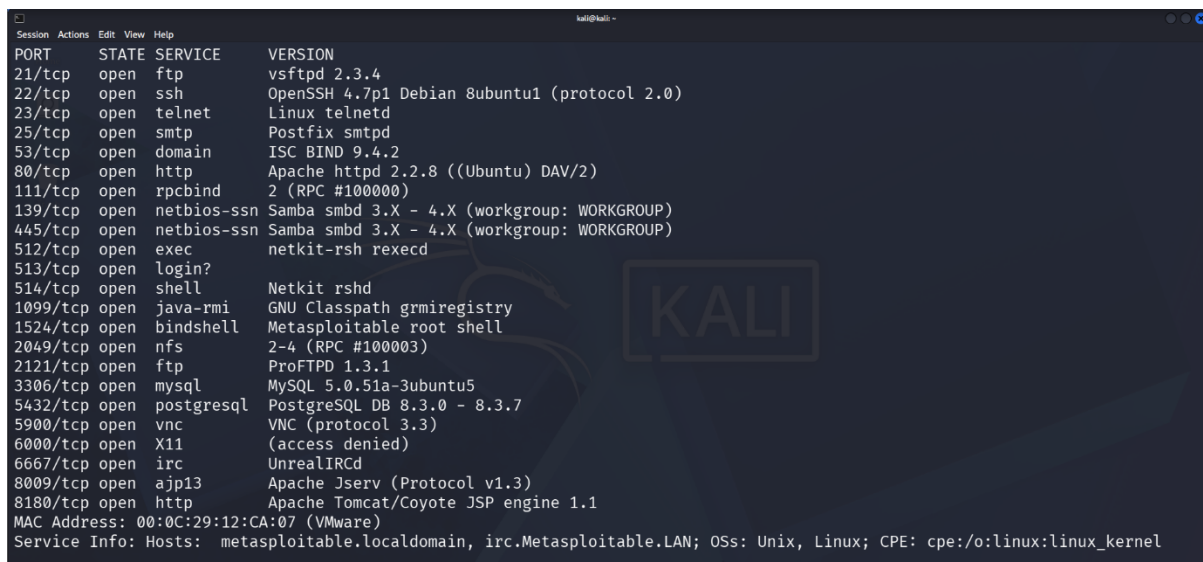
Nmap service detection was performed using:

```
nmap -sV 192.168.47.133
```

The scan identified multiple exposed services, including:

- FTP
- SSH
- Telnet
- SMTP
- DNS
- HTTP
- SMB
- RMI
- NFS
- MySQL
- PostgreSQL
- VNC
- IRC
- Apache Tomcat

The presence of multiple outdated services indicated a large attack surface.



```
Session Actions Edit View Help
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login?       Netkit rshd
514/tcp   open  shell        GNU Classpath grmiregistry
1099/tcp  open  java-rmi     Metasploitable root shell
1524/tcp  open  bindshell    2-4 (RPC #100003)
2049/tcp  open  nfs          ProFTPD 1.3.1
2121/tcp  open  ftp          MySQL 5.0.51a-3ubuntu5
3306/tcp  open  mysql        PostgreSQL DB 8.3.0 - 8.3.7
5432/tcp  open  postgresql   VNC (protocol 3.3)
5900/tcp  open  vnc          (access denied)
6000/tcp  open  X11          UnrealIRCd
6667/tcp  open  irc          Apache Jserv (Protocol v1.3)
8009/tcp  open  ajp13        Apache Tomcat/Coyote JSP engine 1.1
8180/tcp  open  http         MAC Address: 00:0C:29:12:CA:07 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 1: Nmap service and version enumeration of the Metasploitable target

4.3 Vulnerability Scanning

A vulnerability-focused Nmap scan was then performed:

```
nmap -sV --script vuln 192.168.47.133 -oN vuln_scan.txt
```

The scan took approximately 32 minutes and identified multiple known vulnerabilities and potentially insecure configurations.

The results were saved to `vuln_scan.txt` for further analysis.

5. Identified Vulnerabilities

5.1 vsFTPD 2.3.4 Backdoor

Description

The FTP service running on port 21 was identified as:

```
vsftpd 2.3.4
```

The Nmap vulnerability script identified the well-known vsFTPD 2.3.4 backdoor vulnerability.

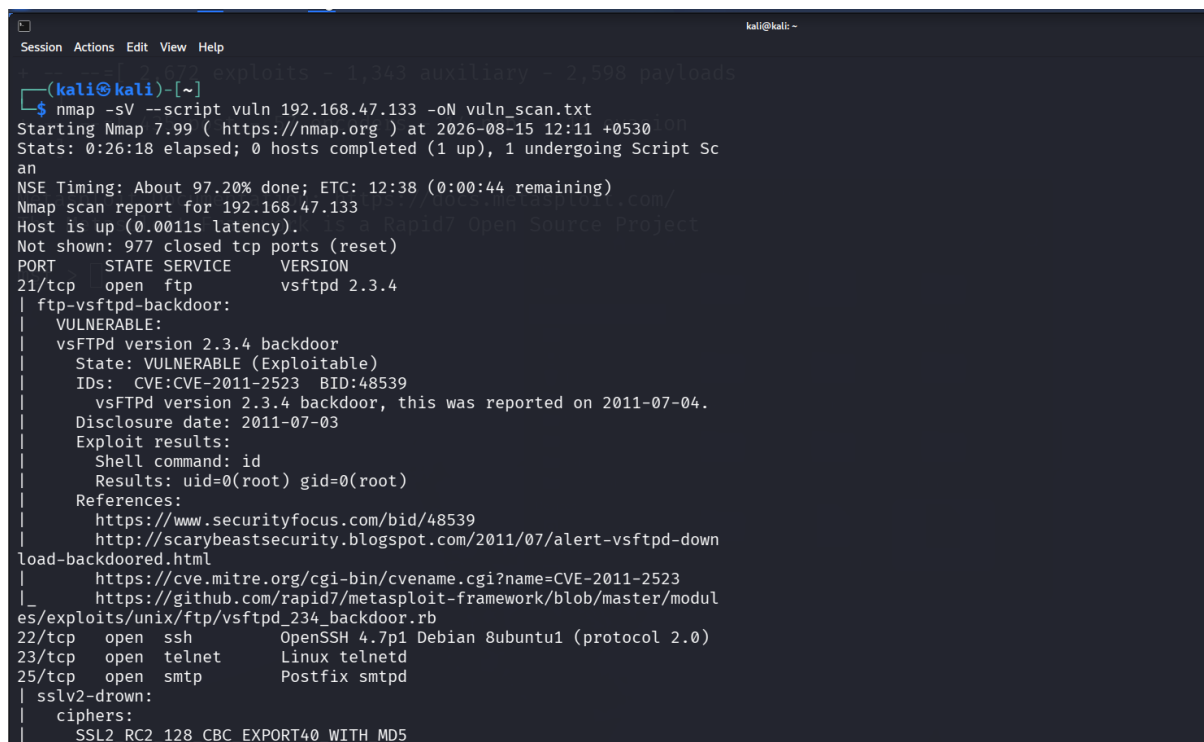
The scan reported:

State: VULNERABLE (Exploitable)
CVE-2011-2523

The NSE check also reported the execution of the `id` command with:

```
uid=0(root) gid=0(root)
```

This indicates an extremely serious vulnerability because successful exploitation can potentially result in unauthorized command execution with root-level privileges.



```
(kali㉿kali)-[~]
└─$ nmap -sV --script vuln 192.168.47.133 -oN vuln_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-15 12:11 +0530
Stats: 0:26:18 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.20% done; ETC: 12:38 (0:00:44 remaining)
Nmap scan report for 192.168.47.133
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPD version 2.3.4 backdoor
|   State: VULNERABLE (Exploitable)
|   IDs: CVE:CVE-2011-2523  BID:48539
|   vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|   Disclosure date: 2011-07-03
|   Exploit results:
|   Shell command: id
|   Results: uid=0(root) gid=0(root)
|   References:
|   https://www.securityfocus.com/bid/48539
|   http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-down
load-backdoored.html
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|   https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
| sslv2-drown:
|   ciphers:
|   SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
```

Figure 2: Nmap detection of the vsFTPD 2.3.4 backdoor vulnerability

Impact

Successful exploitation could allow an attacker to:

- Execute commands on the target system.
- Obtain unauthorized access.
- Execute commands with elevated privileges.
- Compromise the confidentiality, integrity, and availability of the system.

Severity

Critical

Mitigation

- Upgrade vsFTPD to a supported and secure version.
- Remove the vulnerable version from production systems.
- Restrict FTP access using firewall rules.
- Prefer secure alternatives such as SFTP where appropriate.
- Monitor FTP services for suspicious activity.

5.2 Java RMI Registry Remote Class Loading Vulnerability

Description

Port 1099 was identified as running:

```
GNU Classpath grmiregistry
```

Nmap's `rmi-vuln-classloader` script identified the RMI registry as vulnerable to remote class loading.

The scan reported:

```
VULNERABLE
RMI registry default configuration remote code execution vulnerability
```

The vulnerability results indicated that the default configuration allowed classes to be loaded from remote URLs, which can potentially lead to remote code execution.

```
kali@kali: ~  
Session Actions Edit View Help  
1099/tcp open  java-rmi      GNU Classpath grmiregistry  
| rmi-vuln-classloader:  
|   VULNERABLE:  
|   RMI registry default configuration remote code execution vulnerabili  
ty  
|   State: VULNERABLE  
|   Default configuration of RMI registry allows loading classes fro  
m remote URLs which can lead to remote code execution.  
|  
|   References:  
|   https://github.com/rapid7/metasploit-framework/blob/master/modul  
es/exploits/multi/misc/java_rmi_server.rb  
1524/tcp open  bindshell    Metasploitable root shell  
2049/tcp open  nfs          2-4 (RPC #100003)  
2121/tcp open  ftp          ProFTPD 1.3.1  
3306/tcp open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
| ssl-dh-params:  
|   VULNERABLE:  
|   Diffie-Hellman Key Exchange Insufficient Group Strength  
|   State: VULNERABLE  
|   Transport Layer Security (TLS) services that use Diffie-Hellman  
groups  
|   of insufficient strength, especially those using one of a few co  
mmonly  
|   shared groups, may be susceptible to passive eavesdropping attac  
ks.  
|   Check results:  
|   WEAK DH GROUP 1  
|   Cipher Suite: TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA  
|   Modulus Type: Safe prime  
|   Modulus Source: Unknown/Custom-generated  
|   Modulus Length: 1024  
|   Generator Length: 8  
|   Public Key Length: 1024
```

Figure 3: Nmap detection of the Java RMI registry class-loading vulnerability

Impact

An attacker able to reach the RMI service could potentially abuse the insecure configuration to:

- Load unauthorized classes.
- Execute malicious code remotely.
- Gain unauthorized access to the host.
- Compromise the underlying system.

Severity

Critical / High

Mitigation

- Disable RMI if it is not required.
- Restrict access to the RMI registry using firewall rules.
- Disable remote class loading.
- Upgrade vulnerable Java/RMI components.
- Avoid exposing RMI services directly to untrusted networks.

5.3 Web Server and Apache Vulnerabilities

The HTTP service on port 80 was identified as:

Apache httpd 2.2.8 (Ubuntu) DAV/2

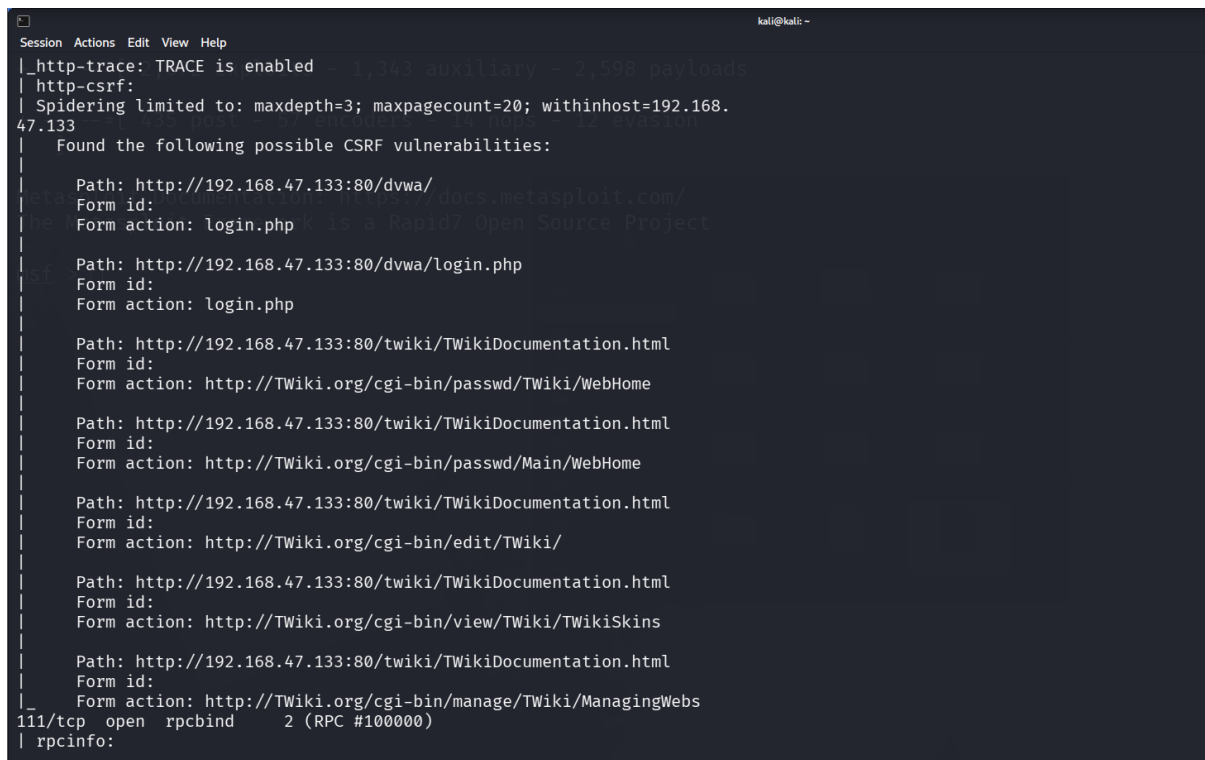
The vulnerability scan identified several security concerns associated with the web server.

These included:

- Slowloris denial-of-service condition — reported as **LIKELY VULNERABLE**.
- HTTP TRACE method enabled.
- Potentially interesting directories.
- PHP information disclosure.
- Exposed phpMyAdmin.
- Test and documentation directories.
- Directory listing exposure.

The scan also identified:

```
/phpinfo.php  
/phpMyAdmin/  
/test/  
/doc/  
/icons/  
/index/
```



```
kali@kali: ~  
Session Actions Edit View Help  
|_ http-trace: TRACE is enabled - 1,343 auxiliary - 2,598 payloads  
|_ http-csrf:  
|_ Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.  
47.133  
|_ Found the following possible CSRF vulnerabilities:  
|  
|   Path: http://192.168.47.133:80/dvwa/  
|   Form id:  
|   Form action: login.php  
|  
|   Path: http://192.168.47.133:80/dvwa/login.php  
|   Form id:  
|   Form action: login.php  
|  
|   Path: http://192.168.47.133:80/twiki/TWikiDocumentation.html  
|   Form id:  
|   Form action: http://TWiki.org/cgi-bin/passwd/TWiki/WebHome  
|  
|   Path: http://192.168.47.133:80/twiki/TWikiDocumentation.html  
|   Form id:  
|   Form action: http://TWiki.org/cgi-bin/passwd/Main/WebHome  
|  
|   Path: http://192.168.47.133:80/twiki/TWikiDocumentation.html  
|   Form id:  
|   Form action: http://TWiki.org/cgi-bin/edit/TWiki/  
|  
|   Path: http://192.168.47.133:80/twiki/TWikiDocumentation.html  
|   Form id:  
|   Form action: http://TWiki.org/cgi-bin/view/TWiki/TWikiSkins  
|  
|   Path: http://192.168.47.133:80/twiki/TWikiDocumentation.html  
|   Form id:  
|   Form action: http://TWiki.org/cgi-bin/manage/TWiki/ManagingWebs  
|_  
111/tcp open  rpcbind      2 (RPC #100000)  
|_ rpcinfo:
```

Figure 4: Nmap web-service vulnerability and enumeration findings

Additional portions of the web-related results were captured separately because the complete output could not fit into a single screenshot.

```
kali@kali: ~  
Session Actions Edit View Help  
| http-slowloris-check: loits ~ 1,343 auxiliary ~ 2,598 payloads  
| VULNERABLE:  
| Slowloris DOS attack  
| State: LIKELY VULNERABLE ncoders ~ 14 nops ~ 12 evasion  
| IDs: CVE:CVE-2007-6750  
| Slowloris tries to keep many connections to the target web serve  
r open and hold  
| them open as long as possible. It accomplishes this by opening  
connections to the target web server and sending a partial request. By doing so  
| it starves  
| the http server's resources causing Denial Of Service.  
|  
| Disclosure date: 2009-09-17  
| References:  
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750  
| http://ha.ckers.org/slowloris/  
|_ http-dombased-xss: Couldn't find any DOM based XSS.  
|_ http-sql-injection: ERROR: Script execution failed (use -d to debug)  
|_ http-enum:  
| /tikiwiki/: Tikiwiki  
| /test/: Test page  
| /phpinfo.php: Possible information file  
| /phpMyAdmin/: phpMyAdmin  
| /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8  
(ubuntu) dav/2'  
| /icons/: Potentially interesting folder w/ directory listing  
|_ /index/: Potentially interesting folder  
|_ http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to d  
ebug)  
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.  
|_ http-trace: TRACE is enabled  
|_ http-csrf:  
| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.  
47.133
```

Figure 5: Additional HTTP security findings identified during vulnerability scanning

Impact

These weaknesses could provide attackers with useful information about the target and may increase the attack surface.

Potential consequences include:

- Denial of service.
- Server information disclosure.
- Exposure of administrative interfaces.
- Exposure of sensitive configuration information.
- Easier reconnaissance for further attacks.

Severity

High

Mitigation

- Upgrade Apache to a currently supported version.
- Disable unnecessary HTTP methods such as TRACE.
- Remove unnecessary test and documentation files.
- Restrict access to administrative interfaces such as phpMyAdmin.
- Disable directory listing where it is not required.
- Remove publicly accessible phpinfo.php.
- Implement appropriate request and connection limits.
- Use a reverse proxy or web application firewall where appropriate.

5.4 SSL/TLS — DROWN Vulnerability

The vulnerability scan identified SSLv2 support associated with the SMTP service.

The Nmap output reported:

```
CVE-2016-0800
OpenSSL: Cross-protocol attack on TLS using SSLv2 (DROWN)
State: VULNERABLE
```



```
Session Actions Edit View Help
| sslv2-drown:
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|   vulns:
|     CVE-2016-0800:
|       title: OpenSSL: Cross-protocol attack on TLS using SSLv2 (DROWN)
|       state: VULNERABLE
|       ids:
|         CVE:CVE-2016-0800
|       description:
|         The SSLv2 protocol, as used in OpenSSL before 1.0.1s and
|         1.0.2 before 1.0.2g and
|         other products, requires a server to send a ServerVerify message
|         before establishing
|         that a client possesses certain plaintext RSA data, which makes
|         it easier for remote
|         attackers to decrypt TLS ciphertext data by leveraging a Bleiche
|         nbacher RSA padding
|         oracle, aka a "DROWN" attack.
```

Figure 6: DROWN vulnerability detected on the target system

Impact

Weak legacy cryptographic protocols can allow attackers to exploit weaknesses in SSL/TLS implementations and potentially recover sensitive information under suitable conditions.

Severity

High

Mitigation

- Disable SSLv2.
- Disable other obsolete SSL/TLS protocols.
- Upgrade OpenSSL and affected services.
- Configure services to use modern TLS versions and secure cipher suites.
- Regularly review cryptographic configurations.

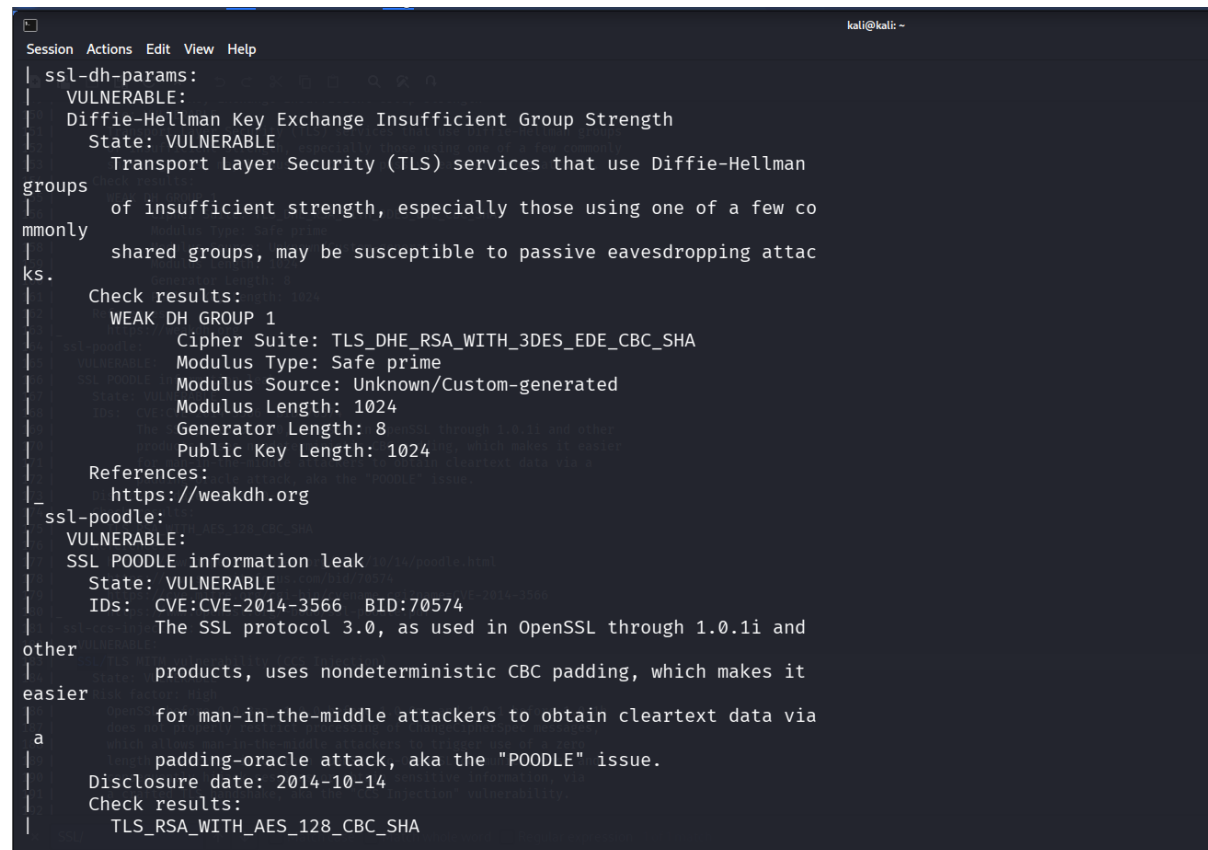
5.5 Additional SSL/TLS Weaknesses

The scan also identified several additional cryptographic weaknesses, including:

- Weak Diffie-Hellman group.
- SSL POODLE vulnerability.
- OpenSSL CCS Injection vulnerability.

The weak DH finding identified a **1024-bit Diffie-Hellman group**, while the POODLE finding was associated with SSL 3.0.

The CCS Injection finding was associated with vulnerable versions of OpenSSL.

A screenshot of a terminal window displaying Nmap scan results for SSL/TLS vulnerabilities. The output is organized into sections: 'ssl-dh-params' showing a 'VULNERABLE' state for 'Diffie-Hellman Key Exchange Insufficient Group Strength' with a 'WEAK DH GROUP 1' and details like 'Modulus Length: 1024'; 'ssl-poodle' showing a 'VULNERABLE' state for 'SSL POODLE information leak' with IDs 'CVE:CVE-2014-3566' and 'BID:70574'; and 'other' showing 'TLS_RSA_WITH_AES_128_CBC_SHA'. The terminal has a dark background with light-colored text.

```
Session Actions Edit View Help
| ssl-dh-params:
|   VULNERABLE:
|   Diffie-Hellman Key Exchange Insufficient Group Strength
|   State: VULNERABLE
|   Transport Layer Security (TLS) services that use Diffie-Hellman
groups
|   of insufficient strength, especially those using one of a few co
mmonly
|   shared groups, may be susceptible to passive eavesdropping attac
ks.
|   Check results:
|   WEAK DH GROUP 1
|   Cipher Suite: TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
|   Modulus Type: Safe prime
|   Modulus Source: Unknown/Custom-generated
|   Modulus Length: 1024
|   Generator Length: 8
|   Public Key Length: 1024
|   References:
|   https://weakdh.org
| ssl-poodle:
|   VULNERABLE:
|   SSL POODLE information leak
|   State: VULNERABLE
|   IDs: CVE:CVE-2014-3566 BID:70574
|   The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and
other
|   products, uses nondeterministic CBC padding, which makes it
easier
|   for man-in-the-middle attackers to obtain cleartext data via
a
|   padding-oracle attack, aka the "POODLE" issue.
|   Disclosure date: 2014-10-14
|   Check results:
|   TLS_RSA_WITH_AES_128_CBC_SHA
```

Figure 7: Additional SSL/TLS vulnerabilities identified by Nmap

```
Session Actions Edit View Help
| ssl-ccs-injection:
|   VULNERABLE:
|     SSL/TLS MITM vulnerability (CCS Injection)
|     State: VULNERABLE
|     Risk factor: High
|     OpenSSL before 0.9.8za, 1.0.0 before 1.0.0m, and 1.0.1 before 1.
0.1h
|     does not properly restrict processing of ChangeCipherSpec messag
es,
|     which allows man-in-the-middle attackers to trigger use of a zer
o
|     length master key in certain OpenSSL-to-OpenSSL communications,
and
|     consequently hijack sessions or obtain sensitive information, vi
a
|     a crafted TLS handshake, aka the "CCS Injection" vulnerability.
|
|   References:
|     http://www.cvedetails.com/cve/2014-0224
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-0224
|     http://www.openssl.org/news/secadv_20140605.txt
```

Figure 8: SSL CSS injection vulnerability identified by Nmap

Impact

These weaknesses may allow attackers to take advantage of outdated cryptographic protocols or weak cryptographic parameters.

Depending on the affected service and attack conditions, consequences may include:

- Information disclosure.
- Weakening of encrypted communications.
- Potential man-in-the-middle attacks.
- Compromise of confidentiality.

Severity

High

Mitigation

- Upgrade OpenSSL and affected services.
- Disable SSLv2 and SSLv3.
- Use modern TLS configurations.
- Replace weak DH parameters with appropriately strong groups.
- Regularly review supported protocols and cipher suites.

6. Manual HTTP Verification

After the automated vulnerability scan, selected web-server findings were manually inspected.

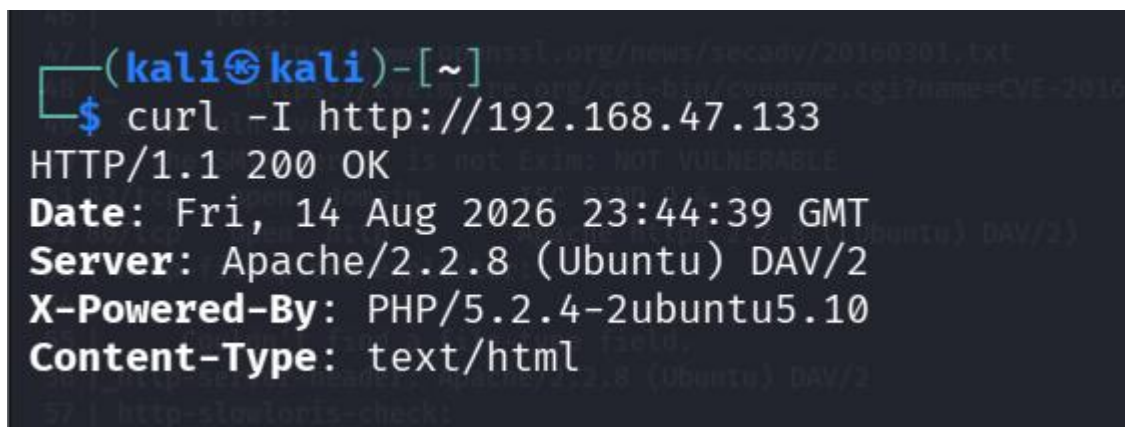
The following command was used:

```
curl -I http://192.168.47.133
```

The server returned:

```
HTTP/1.1 200 OK
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
```

This confirmed that the web server exposes its Apache and PHP versions through HTTP response headers.



```
(kali@kali)-[~]
$ curl -I http://192.168.47.133
HTTP/1.1 200 OK
Date: Fri, 14 Aug 2026 23:44:39 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
```

Figure 9: Manual HTTP header verification using curl

The following Nmap command was also used:

```
nmap -p 80 --script http-headers,http-methods 192.168.47.133
```

The scan reported the supported methods as:

```
GET HEAD POST OPTIONS
```

The results provided additional information about the HTTP configuration.

It should be noted that the presence of `OPTIONS` alone was **not treated as a vulnerability**. The earlier vulnerability scan separately identified that HTTP TRACE was enabled.

7. Vulnerability Summary

Vulnerability	Port	Evidence	Severity
vsFTPD 2.3.4 Backdoor	21	CVE-2011-2523, exploitable, root-level result	Critical
Java RMI Classloader	1099	rmi-vuln-classloader	Critical/High
Slowloris	80	Nmap reported likely vulnerable	High
DROWN / SSLv2	25	CVE-2016-0800	High
SSL POODLE	5432*	CVE-2014-3566	High
CCS Injection	5432*	CVE-2014-0224	High
Weak DH Parameters	5432*	1024-bit DH group	High
HTTP TRACE Enabled	80	Nmap http-trace	Medium
PHP Version Disclosure	80	X-Powered-By header	Medium
Exposed phpMyAdmin/phpinfo	80	http-enum	Medium
Directory/Test Resources	80	http-enum	Medium

*The exact service association should be described as reported by the Nmap scan rather than implying that all TLS findings necessarily represent the same application-layer service.

8. Risk Analysis

The assessment demonstrates that the target has a significantly exposed attack surface.

The most serious finding was the **vsFTPD 2.3.4 backdoor**, which Nmap explicitly classified as exploitable and reported a root-level command execution result.

The **Java RMI class-loading vulnerability** also represents a serious risk because insecure remote class loading can potentially result in remote code execution.

The outdated Apache/PHP environment and multiple SSL/TLS weaknesses further increase the overall security risk.

Because Metasploitable is intentionally designed to contain vulnerabilities, these results are expected within the laboratory environment. However, they demonstrate the types of weaknesses that should be identified and remediated in real systems.

9. Recommended Security Improvements

Based on the assessment, the following actions are recommended:

1. **Upgrade outdated software and services** to supported versions.
2. **Remove unnecessary services** from systems where they are not required.
3. **Restrict exposed services using firewall rules.**
4. **Disable insecure legacy protocols**, including SSLv2 and SSLv3.
5. **Use strong cryptographic parameters** and modern TLS configurations.
6. **Disable unnecessary HTTP methods**, particularly TRACE.
7. **Remove information-disclosure files** such as `phpinfo.php`.
8. **Restrict administrative interfaces** such as phpMyAdmin.
9. **Disable unnecessary directory listings.**
10. **Monitor exposed services and maintain regular vulnerability assessments.**
11. **Apply security patches regularly.**
12. **Use network segmentation** to prevent vulnerable services from being directly accessible from untrusted networks.

10. Limitations

This assessment was performed in a controlled laboratory environment against an intentionally vulnerable Metasploitable virtual machine.

The assessment primarily focused on:

- Network reconnaissance.
- Service enumeration.
- Nmap vulnerability detection.
- Limited manual verification.

Not every vulnerability identified by Nmap was manually exploited. Therefore, vulnerability scanner results such as **"LIKELY VULNERABLE"** should be treated as findings requiring further validation rather than proof of successful exploitation.

No attacks were performed against external or unauthorized systems.

11. Conclusion

The vulnerability assessment successfully identified multiple security weaknesses in the Metasploitable test network.

The assessment discovered numerous outdated and vulnerable services, including the **vsFTPD 2.3.4 backdoor**, **Java RMI remote class-loading vulnerability**, outdated

Apache/PHP components, SSL/TLS weaknesses, HTTP configuration issues, and information-disclosure risks.

The results demonstrate the importance of maintaining updated software, minimizing exposed services, securing network access, and regularly performing vulnerability assessments.

The exercise provided practical experience in reconnaissance, service enumeration, automated vulnerability scanning, manual verification, vulnerability classification, and security remediation.